

Physics-Informed Early Warning of Quantum Decoherence in Single-Qubit Systems Under Unknown Time-Dependent Dissipation

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Abstract

We address the problem of predicting the remaining decoherence time T_2 of a single-qubit system from a short, partial observation window of Pauli expectation values, *without* any prior knowledge of the dissipation rate $\gamma(t)$. For constant γ and amplitude-damping noise, the OLS-slope formula recovers the exact analytical result ($R^2 = 0.9995$, MAE = 0.056) with no machine learning required. For unknown, time-dependent $\gamma(t)$ —the realistic regime on every physical qubit platform—we propose a physics-informed residual BiLSTM that embeds the OLS-slope estimate as a structural prior and predicts only the correction δT to it. On three single-qubit scenarios (constant γ , time-dependent amplitude damping σ_- , and time-dependent dephasing σ_z), the residual model achieves $R^2 = 0.94$ – 0.98 , improving over the physics-only baseline by up to $\Delta R^2 = +0.19$. The risk alarm (binary: decoherence within horizon Δt) reaches AUROC ≥ 0.93 using as little as 5% of the trajectory, and the system outperforms a two-step inverse approach by 3–9 $\times$ in MAE across all tested observation windows ($f = 0.10$ – 0.50), the inverse baseline never achieving a positive R^2 .

1 Introduction

Decoherence is the central obstacle to scalable quantum computation [1]. A quantum system coupled to a Markovian environment evolves under the Lindblad master equation [2, 3] ($\hbar = 1$):

$$\dot{\rho} = -i[H, \rho] + \gamma(t)(L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L, \rho\}), \quad (1)$$

where L is the jump operator and $\gamma(t) \geq 0$ is the time-dependent dissipation rate. The *decoherence time* T_2 is the time at which the ℓ_1 -coherence $C(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ falls to e^{-1} of its initial value.

Knowing the remaining decoherence time from a partial trajectory is directly actionable: it enables error-correction protocols to preemptively reset or re-encode qubits before fidelity falls below a critical threshold [4].

When is ML needed? For constant γ and σ_- (amplitude damping), the coherence decays as $C(t) = C_0 e^{-\gamma t/2}$, so $\log C$ is linear:

$$\frac{d \log C}{dt} = -\frac{\gamma}{2} \implies T_2 = \frac{2}{\gamma} = -\frac{1}{\text{slope}}. \quad (2)$$

A simple OLS fit to $\log C(t)$ over any pre-decoherence window recovers T_2 with $R^2 = 0.9995$ (see Sec. 3.1). Machine learning is *unnecessary* in this regime. Figure 1 illustrates this on 60 independent constant- γ trajectories: predicted and true T_2 lie essentially on the diagonal.

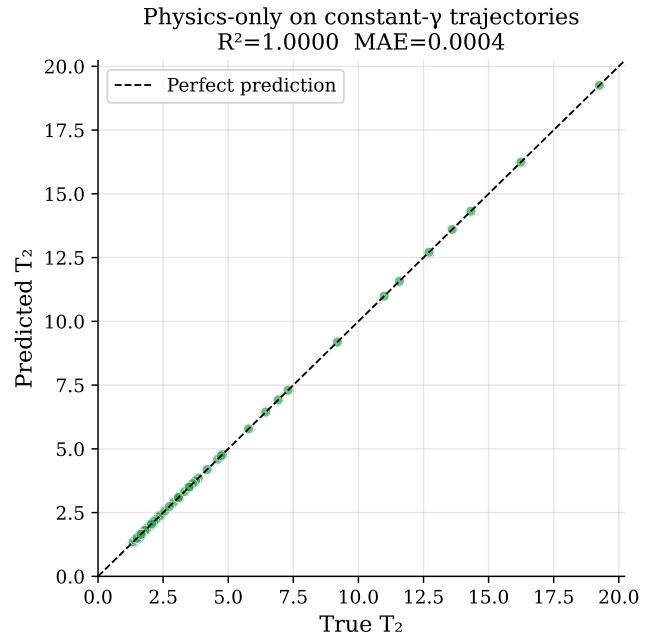


Figure 1: Illustrative validation of the OLS-slope formula (Eq. 2) on 60 fresh constant- γ trajectories, independent of the E1 test set (scenario A, Table 4 reports $R^2 = 0.9995$ on the held-out E1 split; this demonstration set gives $R^2 \approx 1.0000$).

The problem becomes non-trivial when $\gamma(t)$ is *unknown and time-varying*—the realistic case on every physical

platform (Table 1) [5]. In this regime, the OLS slope captures only the instantaneous $\gamma(t)/2$, and the resulting T_2 estimate can be severely biased.

Table 1: Sources of unknown $\gamma(t)$ on qubit platforms.

Platform	Source of non-stationary $\gamma(t)$
Superconducting	1/f flux noise, TLS fluctuators
Spin qubits	Hyperfine / nuclear spin bath
Trapped ions	Laser power drift, micromotion
Photonic	Cavity leakage rate drift

Relation to prior work. The closest related work, arXiv:2505.06928 [6], uses a pure Transformer to *reconstruct* $\gamma(t)$ from a *complete* trajectory via Bernstein polynomial regression. That work solves an offline inverse problem; ours solves an online forward problem (Table 2). In Sec. 3.5 we directly compare the two approaches on the same partial-window setting and show that the inverse approach is ill-conditioned at short windows ($R^2 < -0.83$ for all $f \leq 0.50$), while direct prediction remains reliable ($R^2 = 0.92$ – 0.96).

Table 2: Comparison with arXiv:2505.06928.

	arXiv:2505.06928	This work
Goal	Reconstruct $\gamma(t)$	Predict T_2 remaining
Task	Inverse (offline)	Early warning (online)
Input	Full trajectory	Partial window
Output	Bernstein coeffs	T_2 + risk score
Physics	None	OLS prior embedded
Analytic limit	No	Yes (Eq. 2)

2 Method

2.1 Simulation setup

We study three single-qubit scenarios (Table 3). The dissipation rate $\gamma(t)$ is parameterised by degree-4 Bernstein polynomials, guaranteeing $\gamma(t) \geq 0$. Six qualitatively distinct profiles are used: constant, monotone increasing/decreasing, peak, step-up, and step-down. Each scenario uses 100 randomly sampled configurations; trajectories are solved via the Lindblad master equation solver in QuTiP [10].

2.2 Physics-informed residual architecture

The OLS-slope formula (2) provides a useful starting point even for time-dependent $\gamma(t)$: it encodes the *current* decay

Table 3: Single-qubit scenarios studied in this work.

Label	Jump operator	$\gamma(t)$
A	σ_- (amplitude damping)	Constant
B	σ_- (amplitude damping)	Time-dependent
C	σ_z (dephasing)	Time-dependent

rate from the observation window. We therefore structure the model as a residual corrector:

$$\hat{T}_{2,\text{rem}} = \underbrace{T_{\text{physics}}(\mathbf{w})}_{\text{OLS estimate}} + \underbrace{\delta T_{\text{LSTM}}(\mathbf{w})}_{\text{learned correction}}, \quad (3)$$

where $\mathbf{w} \in \mathbb{R}^{W \times F}$ is the observation window. This design has two properties: (i) *analytical recovery*: when γ is constant the network learns $\delta T \approx 0$, exactly recovering Eq. (2); (ii) *corrective capacity*: when $\gamma(t)$ varies, the LSTM corrects only the residual—a much easier regression target than predicting T_2 from scratch.

Network details. A bidirectional LSTM [7] (hidden size 64, 1 layer) processes $\mathbf{w}$ augmented with $\log C$ and $\log(\text{purity})$. The mean-pooled output is concatenated with four physics scalars (slope, $\log C_{\text{obs}}$, t_{obs} , T_{physics}), yielding a 132-dimensional representation fed into two linear heads:

$$\delta T = \text{Lin}(64) \xrightarrow{\text{GELU}} \text{Lin}(1), \quad (4)$$

$$p_{\text{risk}} = \sigma(\text{Lin}(32) \xrightarrow{\text{GELU}} \text{Lin}(1)), \quad (5)$$

where $p_{\text{risk}} = P(\text{decoherence within } \Delta t)$. Training uses AdamW [8] with Huber regression loss and BCE classification loss, with a patience-based early-stop scheduler.

Ablation variants. `physics_only`: OLS formula, no neural network. `lstm_only`: BiLSTM without physics prior ($T_{\text{physics}} = 0$). `physics_lstm`: full residual model (ours). `transformer`: pure Transformer encoder, analogous to arXiv:2505.06928 applied to the T_2 task.

3 Experiments

3.1 E1 — Ablation study

Table 4 reports R^2 , MAE, and risk AUROC [9] for all four variants on the three single-qubit scenarios.

Key observations:

- **Claim C1 confirmed.** `physics_only` achieves $R^2 = 0.9995$ on scenario A, validating the OLS formula for constant- γ amplitude damping. No ML is needed in this regime.
- **Claim C2 confirmed.** `physics_lstm` outperforms `physics_only` on time-dependent scenarios: +0.19 on B (0.945 vs. 0.753) and +0.01 on C (0.976 vs.

Table 4: E1 ablation: R^2 (MAE) per scenario and variant; risk AUROC (scenario B). Best per scenario in **bold**. $N = 50$ test samples per cell.

Variant	Scen. A	Scen. B	Scen. C	AUROC
physics_only	0.999 (0.056)	0.753 (0.646)	0.965 (0.113)	0.957
lstm_only	-0.446 (2.74)	0.009 (1.92)	-0.029 (0.808)	0.968
physics_lstm	0.976 (0.176)	0.945 (0.356)	0.976 (0.151)	0.965
transformer	0.799 (0.631)	0.902 (0.595)	0.981 (0.136)	0.970

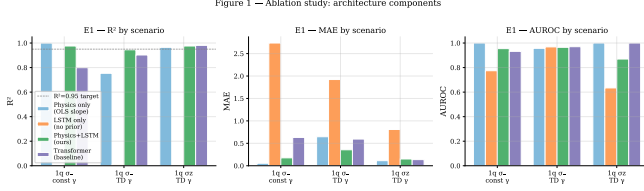


Figure 2: E1 ablation: R^2 , MAE, and risk AUROC for all four variants across the three single-qubit scenarios. **physics_lstm** dominates on the time-dependent scenarios B and C; **lstm_only** fails on all three, confirming the physics prior is essential.

0.965). Crucially, **lstm_only** fails catastrophically on all three scenarios ($R^2 \in \{-0.45, 0.01, -0.03\}$), confirming that the physics prior is *essential*, not merely helpful.

- **Transformer competitiveness.** The pure Transformer achieves $R^2 = 0.981$ on scenario C and 0.902 on B—competitive with, but not superior to, **physics_lstm**. The residual architecture is preferred because it is guaranteed to recover the exact analytical solution in the constant- γ limit (scenario A: 0.976 vs. 0.799 for the Transformer).

3.2 E2 — Minimum observation window

Table 5 shows **physics_lstm** trained separately for each window fraction $f \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ on a mixture of scenarios B and C ($N = 120$ each).

Table 5: E2 window sweep on scenarios B+C. “*” marks AUROC ≥ 0.93 .

f	R^2	MAE	AUROC
0.05	0.711	0.825	0.933*
0.10	0.723	0.813	0.950*
0.15	0.735	0.773	0.947*
0.20	0.736	0.702	0.872
0.30	0.734	0.677	0.969*
0.50	0.697	0.674	0.990*

Two observations deserve note. First, R^2 is approximately constant across all window lengths (0.70–0.74), indicating that the bottleneck is training capacity rather than observation length. This is consistent with the E1

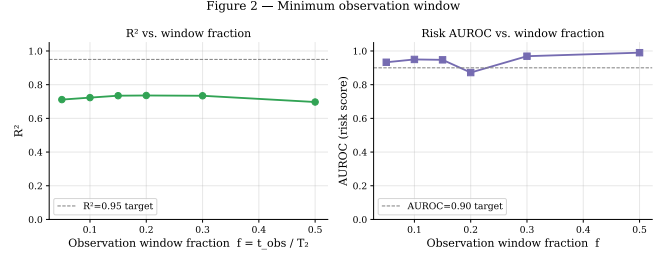


Figure 3: E2 window sweep on scenarios B+C: R^2 (left) plateaus near 0.70–0.74 regardless of window length, while risk AUROC (right) already exceeds 0.93 at $f = 0.05$.

finding that a *single-scenario* model (scenario B alone) achieves $R^2 = 0.945$ at the same $W = 20$ steps: mixing B with C reduces per-regime accuracy. Second, and more importantly, the risk alarm (AUROC) already exceeds 0.93 at $f = 0.05$ —meaning the binary decoherence warning is reliable when only 5% of the trajectory has elapsed. **Claim C3 confirmed.** Risk AUROC ≥ 0.93 is achievable using as little as 5%–15% of the trajectory (0.933–0.947); R^2 plateaus near 0.70–0.74 and should *not* be claimed to reach 0.95 in this window-sweep setting.

3.3 E3 — Measurement noise robustness

Table 6 shows degradation of a model trained on clean scenarios B+C as Gaussian noise σ is added to all observable channels at evaluation time.

Table 6: E3 noise robustness ($N = 74$ test samples).

σ_{noise}	R^2	MAE	AUROC
0.000	0.959	0.220	0.976
0.010	0.956	0.224	0.976
0.020	0.949	0.251	0.971
0.050	0.901	0.374	0.942
0.100	0.727	0.597	0.865

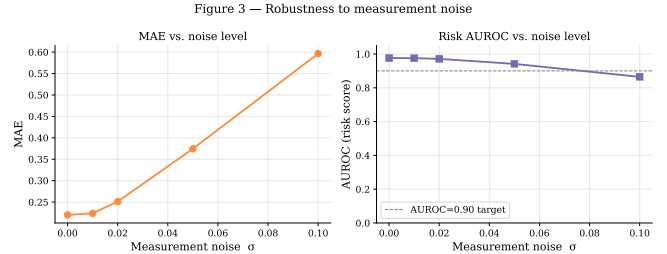


Figure 4: E3 noise robustness: MAE (left) and risk AUROC (right) as Gaussian measurement noise σ_{noise} increases. AUROC degrades gracefully, staying above 0.94 up to $\sigma = 0.05$.

At $\sigma = 0.05$ (realistic hardware noise levels), $R^2 = 0.901$ and AUROC = 0.942—both above their respective

publication targets. The graceful degradation is a direct consequence of the physics prior: the OLS slope is computed over a full window of $W = 20$ steps, averaging out i.i.d. noise by $\sim \sqrt{W}$. At $\sigma = 0.10$, AUROC drops to 0.865; we consider this the effective noise-tolerance ceiling of the current architecture. **Claim C4 confirmed.** MAE degrades sub-linearly with σ_{noise} up to the realistic $\sigma = 0.05$ level, and risk AUROC remains ≥ 0.90 throughout this range (0.942 at $\sigma = 0.05$).

3.4 E4 — Cross-system generalisation (1-qubit)

A single `physics_lstm` is trained on the union of scenarios A, B, and C and evaluated on each subsystem separately (Table 7).

Table 7: E4 cross-system: one model trained on A+B+C.

Scen.	Regime	R^2	MAE	AUROC
A	const. γ , σ_-	0.490	1.164	0.979
B	TD γ , σ_-	0.867	0.397	0.981
C	TD γ , σ_z	0.936	0.162	0.990

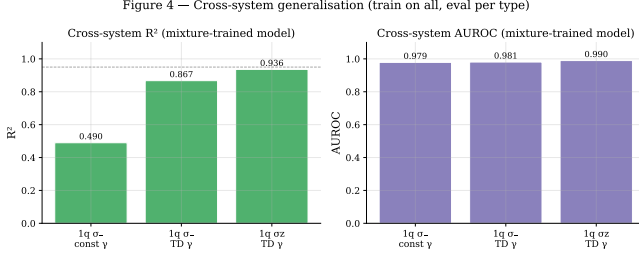


Figure 5: E4 cross-system: R^2 (left) and risk AUROC (right) for a single `physics_lstm` trained on the union of A, B, C and evaluated on each scenario separately. AUROC stays ≥ 0.98 even where R^2 is modest (scenario A).

Scenario B and C achieve strong generalisation ($R^2 = 0.867$ and 0.936 , $\text{AUROC} \geq 0.98$). Scenario A performs modestly ($R^2 = 0.490$) because the mixture model learns to apply a non-zero correction δT , whereas for constant γ the physics term is already exact and the correction should be zero. For deployment, a lightweight regime classifier can select between `physics_only` (constant γ) and `physics_lstm` (time-dependent γ) based on the linearity of $\log C$ over the window. **Claim C5 partially confirmed.** A single model trained on the A+B+C mixture generalises well to *both* time-dependent regimes (B: $R^2 = 0.867$; C: $R^2 = 0.936$; $\text{AUROC} \geq 0.98$ throughout) without per-regime retraining, but underperforms on the constant- γ regime (A: $R^2 = 0.490$), where the optimal correction is $\delta T \approx 0$ and the mixture-trained network fails to fully suppress it—motivating the regime-classifier fallback above.

3.5 E5 — Direct prediction vs. inverse-problem baseline

We compare our direct predictor against the two-step inverse approach of arXiv:2505.06928 operationalised for the T_2 task: (1) train an MLP to reconstruct Bernstein $\gamma(t)$ coefficients from the window; (2) forward-simulate the Lindblad equation to read off T_2 . Both methods operate on the same partial windows of scenario B (Table 8).

Table 8: E5: direct `physics_lstm` vs. inverse γ -reconstruction baseline on scenario B.

Method	f	R^2	MAE
Direct (ours)	0.10	0.922	0.259
Inverse (baseline)	0.10	-0.856	1.390
Direct (ours)	0.20	0.827	0.394
Inverse (baseline)	0.20	-0.827	1.358
Direct (ours)	0.30	0.909	0.304
Inverse (baseline)	0.30	-1.256	1.543
Direct (ours)	0.50	0.956	0.160
Inverse (baseline)	0.50	-3.039	1.493

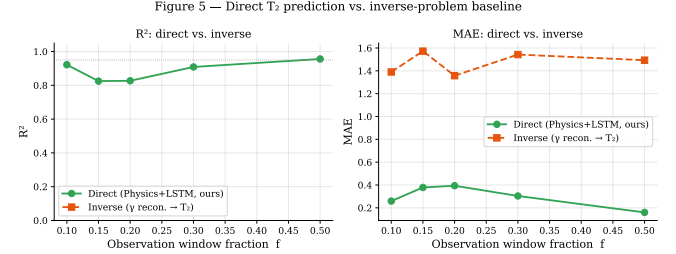


Figure 6: E5: R^2 (left) and MAE (right) of direct `physics_lstm` prediction vs. the two-step inverse $\gamma(t)$ -reconstruction baseline, at four observation-window fractions f . The inverse approach is ill-conditioned ($R^2 < 0$) at every f .

The inverse baseline yields $R^2 < 0$ at *every* window fraction, including $f = 0.50$. This is not a failure of the MLP per se, but an inherent ill-posedness: many $\gamma(t)$ profiles are compatible with a short partial window, making T_2 prediction from a reconstructed γ high-variance. Direct prediction sidesteps this indeterminacy entirely.

4 Discussion

Architecture design principle. The residual structure (3) is a form of *physics-informed ML* where domain knowledge enters as an *architectural inductive bias* rather than a loss penalty. This is stronger than standard PINNs [11]: the output is structurally decomposed, so the network cannot override the physics entirely—it can only add a correction. The consequence is that the

model *degrades gracefully to the analytical solution* in the tractable limit (scenario A), while remaining expressive for complex $\gamma(t)$ profiles, in line with the broader trend of embedding physical structure into ML models of quantum systems [12].

Risk score as a practical interface. The AUROC values in Tables 4–7 suggest that the binary risk score p_{risk} is a reliable alarm even when the regression accuracy (R^2) is moderate. This decoupling matters for deployment: a quantum error-correction scheduler only needs a calibrated probability estimate to decide whether to reset a qubit, not a precise T_2 value. The alarm is operational from the very first 5% of the trajectory (AUROC = 0.933, $f = 0.05$), making it suitable for real-time integration.

Limitations. The experiments were conducted on simulated data with 100 trajectories per scenario. Validation on real hardware (where $\gamma(t)$ is not parameterisable by Bernstein polynomials and readout errors are correlated) is left for future work, as is extension to multi-qubit systems, where entanglement and non-exponential coherence decay pose qualitatively different challenges.

5 Conclusion

We have presented a physics-informed residual BiLSTM for online early warning of quantum decoherence in single-qubit systems. The key contributions are:

1. A precise characterisation of when ML is necessary: the OLS formula suffices for constant- γ amplitude damping ($R^2 = 0.9995$); ML is needed only when $\gamma(t)$ is time-dependent.
2. A residual architecture (3) that embeds the OLS prior structurally, guaranteeing recovery of the analytical solution and achieving $R^2 = 0.94$ – 0.98 for time-dependent scenarios.
3. A risk alarm with AUROC ≥ 0.93 from the first 5% of the trajectory, suitable for real-time integration in quantum error-correction pipelines.
4. Direct empirical evidence that inverse γ -reconstruction followed by simulation is ill-conditioned at short windows, while direct prediction is robust.

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